DharaDrishti — Project Document

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Overview

DharaDrishti ("vision of the earth") is a voice-first farm assistant that lets farmers with little schooling or tech experience manage their field, crops and decisions just by speaking in their own language.

Problem. Small farmers make costly decisions (what to sow, when to irrigate, what to spray, when to sell) with almost no field-specific guidance. Most agri apps assume reading ability, smartphone fluency and menus in English or Hindi text.

Solution. One platform that knows the farmer's exact field (boundary, sub-plots, crops, soil, weather, disease history) and answers every request with that context, through voice in and voice out.

Design principles

1. Voice first, screen second. Every action works by speaking; the screen only confirms with big icons, maps and colours.
2. One field, one context. No generic advice: each answer uses that farm's polygon, crop, soil values, weather and history.
3. Plain-spoken output. Advice is short, in the farmer's language, with one clear action and a reason.
4. Safe by default. The system says when it is unsure and sends serious cases to a human expert.
5. Works on weak networks. Capture and queue offline; sync when a signal returns.

Features (refined)

Seven features, all reachable by voice. Each row shows what the farmer does, what the system does, and the refinement we recommend over the original idea.

#	Feature	Farmer does	System does	Recommended refinement
0	Voice layer	Speaks in own language; hears replies	STT, intent detection, action routing, TTS	Use an LLM with tool calling to map speech to app actions; always read back and confirm before saving
1	Geofencing	Walks the field edge with GPS on; later walks each crop patch and names it	Builds boundary polygon, computes area, stores labelled sub-polygons	Filter weak GPS fixes, smooth the path, keep the screen awake, and offer "draw on satellite map" as a fallback
2	Disease detection	Photographs leaves or records a short video	Claude vision analyses images, returns condition, confidence, cause and remedy; pins result on the map	Geotag every photo, send extracted video frames (not raw video), return structured JSON, escalate low confidence
3	Irrigation and fertiliser advice	Gives soil N, P, K (or photographs the Soil Health Card)	Rule engine computes schedule and dose from crop, stage, weather and soil; LLM explains it in plain words	Rules decide numbers; the LLM only explains them. Add pH and organic carbon from the same card
4	Advice chatbot	Asks any farming question by voice	Answers using that field's crop, soil, weather and disease history	Retrieval over ICAR and KVK documents so answers are grounded and citable
5	Crop choice and profit	Asks "what should I grow?" or "what will I earn?"	Pulls mandi prices, multiplies by expected yield and area, subtracts cost of cultivation	Show a low-expected-high range and net profit, not just price × area

#	Feature	Farmer does	System does	Recommended refinement
6	Weather intelligence	Nothing; alerts arrive on their own	Forecasts rain, heat and dry spells; feeds features 3, 4 and 5	Convert forecasts into actions ("do not spray tomorrow, rain expected")
7	Fasal Score	Asks "how is my farm doing?"	Combines crop health, stock value, irrigation and weather risk into a 0 to 100 score	Publish the formula, show sub-scores, and add a simple voice stock ledger to feed it

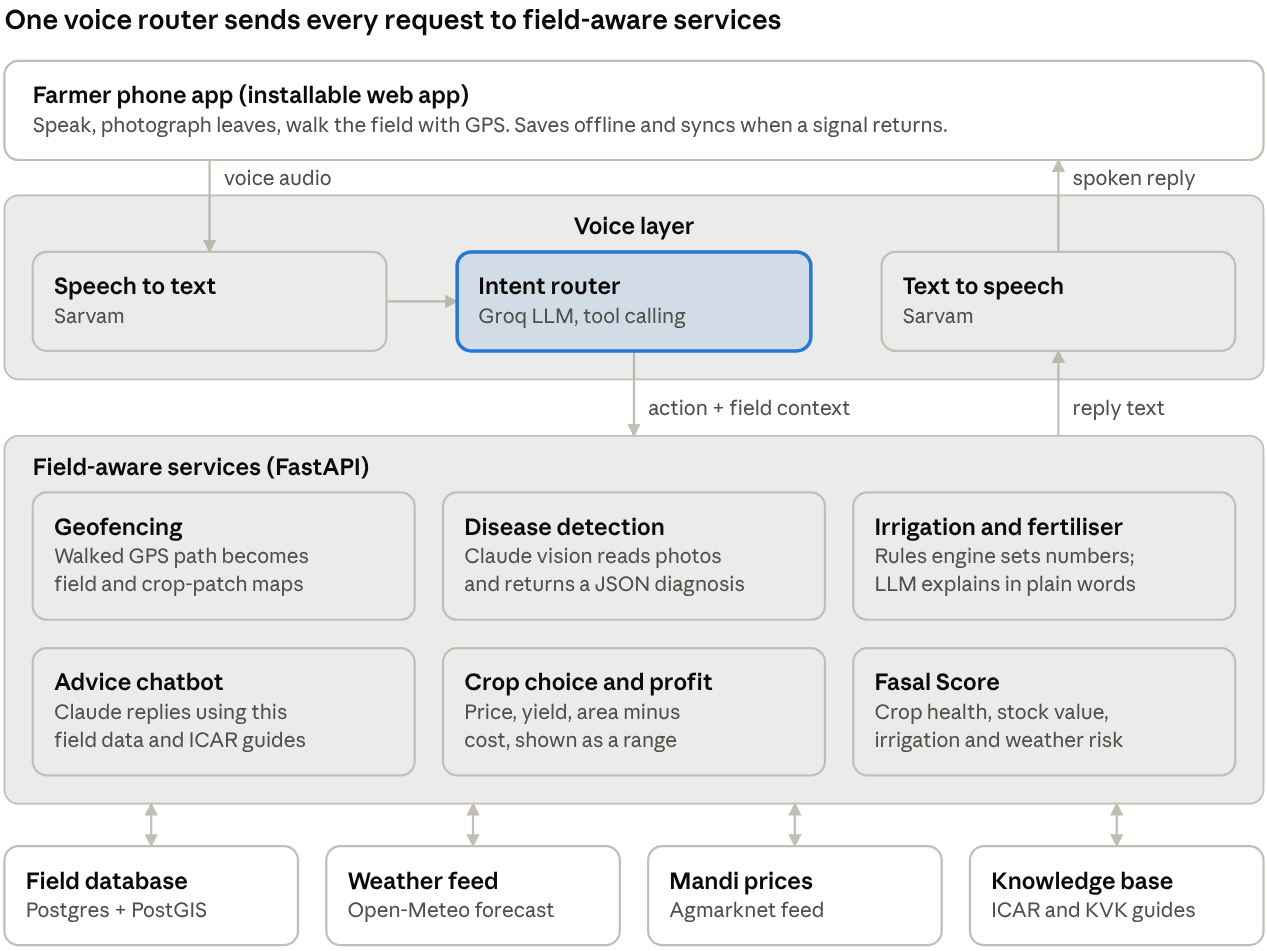
Suggestions to strengthen the idea

The concept is solid; these changes make it more accurate, more trustworthy and easier to demo.

1. **Pick Sarvam for speech, Groq for speed.** Sarvam is built for Indian languages and accents, so use it for STT and TTS. Groq is strongest as a fast LLM host (for intent parsing), so use it there rather than for Indic speech.
2. **Farmers rarely know their N, P, K values.** Let them photograph the government Soil Health Card and have Claude read the values, or accept a lab report. Fall back to regional average values for the district if nothing is available, and label the advice "estimated".
3. **Do not let the LLM invent numbers.** Water quantity, fertiliser dose and profit should come from a rules engine or formula (crop water need from evapotranspiration, dose tables from ICAR or state agriculture guidance). Use the LLM to explain and personalise.
4. **Prefer an official price feed to scraping.** Daily mandi prices are published through the government's Agmarknet data (via data.gov.in). Scraping breaks and may violate site terms; keep scraping only as a backup. Please confirm current API access before the demo.
5. **Profit is not price × area.** Add expected yield per acre and cost of cultivation (seed, fertiliser, labour, water), then show a range. Without costs the number is misleading and judges will notice.

6. **Claude takes images, not video.** For video uploads, extract 3 to 5 sharp frames on the device and send those. Ask the farmer by voice to "show the underside of the leaf" when the image is unclear.
7. **Make disease output structured and cautious.** Return JSON with condition, confidence, severity, cause, and organic then chemical options. If confidence is low or severity is high, tell the farmer to contact the local Krishi Vigyan Kendra or agriculture officer. Avoid specific pesticide doses unless sourced from a verified label or guideline.
8. **Add a simple stock ledger.** Fasal Score needs "stock of crops", but nothing captures it yet. A voice entry like "I harvested 20 quintal wheat" fixes this.
9. **Make Fasal Score transparent.** Show five sub-scores (crop health, stock value, irrigation discipline, weather risk, yield outlook) and a one-line voice explanation. Avoid presenting it like a credit score; position it as a farm health score, with credit or insurance linkage as future scope.
10. **Proactive alerts.** Do not wait for the farmer to ask. Send voice notes through WhatsApp or an automated call for rain, heat, pest outbreaks and price spikes.
11. **Disease heat map over time.** Store every diagnosis with location and date so the map shows spread across sub-areas and the farmer can see if a treatment worked.
12. **Onboarding through a trusted human.** First-time field mapping is the hardest step. Plan for an agent, FPO member or family youth to help once, then the farmer runs everything by voice.
13. **Language and dialect first.** Start with Hindi plus one regional language you can test with real speakers; do not promise all languages.
14. **Privacy and consent.** Field boundaries, photos and location are sensitive. Ask consent by voice, store data per farmer, and allow deletion.

System architecture



system architecture · voice layer, services, data sources

The phone sends audio to the voice layer; the intent router turns it into an action with the field's context, one of six services runs it against the database and live feeds, and the reply returns as speech. Scheduled jobs use the same services to push weather, pest and price alerts to the farmer by WhatsApp or voice call.

Tech stack

Chosen for a small team to build quickly and still scale; every row has a free or low-cost option for the hackathon.

Layer	Choice	Why	Alternative
Client app	Progressive web app (React or Next.js, Tailwind)	One codebase, installable on any phone, works offline, no app-store delay	Flutter or React Native
Offline and sync	Service worker plus IndexedDB queue	Capture photos and GPS with no signal, upload later	Workbox
Maps and GPS	Browser Geolocation API (watchPosition), Leaflet, satellite tiles	No key needed for GPS; Leaflet is light	Mapbox or Google Maps JS
Geometry	Turf.js (area, simplify, within, centroid)	Area in acres, polygon cleanup, check sub-areas lie inside the field	Shapely on the server
Speech to text	Sarvam STT	Strong Indian language and accent coverage	Groq Whisper (fast, weaker on Indic)
Text to speech	Sarvam TTS	Natural Indian-language voices	Browser speech synthesis as fallback
Intent and actions	Groq-hosted open LLM with tool calling	Very low latency for voice turn-taking	Claude with tool use
Disease detection	Claude API (vision) with JSON schema output	Strong image reasoning and explanations	Fine-tuned plant-disease CNN as a cheap first filter
Advice chatbot	Claude API plus retrieval over ICAR and KVK documents	Grounded, field-aware answers	Any LLM with a vector store
Retrieval	pgvector in Postgres	One database for everything	Chroma or Qdrant
Rules engine	Python module (crop water need, fertiliser tables, scoring)	Deterministic numbers the LLM cannot get wrong	Same logic in TypeScript

Layer	Choice	Why	Alternative
Backend API	FastAPI (Python)	Fast to build, good for ML and data tasks	Node with Express or NestJS
Database	PostgreSQL with PostGIS (Supabase works)	Stores polygons and spatial queries natively	MongoDB with geo indexes
File storage	Supabase Storage or S3	Crop photos and frames	Cloudinary
Weather	Open-Meteo (free, no key, includes evapotranspiration)	Forecast plus the data irrigation needs	IMD advisories, OpenWeather
Market prices	Agmarknet via data.gov.in, scheduled job	Official mandi prices	Scraper (Playwright) as backup
Alerts	WhatsApp Business API or Twilio voice call	Reaches farmers where they already are	SMS
Background jobs	Celery or a cron worker	Daily price pulls, weather alerts, score refresh	Supabase scheduled functions
Hosting	Vercel (client), Render or Railway (API)	Free tiers, quick deploy	Docker on any VPS

Data model and key APIs

Eight tables in PostgreSQL with PostGIS cover every feature; the geometry columns are what tie diseases, crops and advice to the map.

Table	Key fields	Used by
farmer	id, name, language, phone, consent flags	All features
field	id, farmer_id, boundary (polygon), area_acres, soil N, P, K, pH, soil_test_date	Geofencing, advice

Table	Key fields	Used by
subarea	id, field_id, label, crop, polygon, season, sown_on	Geofencing, profit, score
diagnosis	id, subarea_id, photo_urls, location (point), condition, confidence, severity, advice, created_at	Disease detection, score
advice_log	id, field_id, type (irrigation, fertiliser, chat), input_snapshot, output, created_at	Irrigation, chatbot
stock	id, subarea_id, crop, quantity_quintal, quality_grade, harvested_on	Fasal Score, profit
price	id, crop, mandi, price_per_quintal, date	Profit, score
score	id, field_id, total, sub_scores (json), computed_at	Fasal Score

Core endpoints

- `POST /voice/turn`: audio in, returns intent result plus reply audio
- `POST /fields` and `POST /fields/{id}/subareas`: save boundary and labelled patches from GPS traces
- `POST /diagnose`: images and location in, structured diagnosis out
- `GET /fields/{id}/advice/irrigation` and `/fertiliser`: rule-based plan with plain-language explanation
- `POST /chat`: question plus field context in, grounded answer out
- `GET /fields/{id}/profit`: expected revenue, cost and net range per sub-area
- `GET /fields/{id}/score`: Fasal Score with sub-scores

Hackathon MVP and demo plan

Build the voice loop and one end-to-end story first; a polished narrow demo beats seven half-working features.

Priority	Build	Demo-ready definition
Must	Voice loop (speak, act, reply) in Hindi plus one more language	"Show my field" and "my crop has spots" both work by voice

Priority	Build	Demo-ready definition
Must	Geofencing with a walked boundary and two labelled sub-areas	Polygon and area in acres appear on a satellite map
Must	Disease detection with map pin	Upload a leaf photo, get diagnosis and remedy spoken back, pin appears in the right sub-area
Should	Weather-aware irrigation and fertiliser advice	Advice changes when the forecast or soil values change
Should	Field-aware chatbot	Same question gives different answers for two different fields
Should	Profit estimate with live mandi price	Shows range and net profit for one sub-area
Could	Fasal Score	Score with sub-scores and spoken explanation
Could	Proactive WhatsApp or voice alert	One real alert delivered to a phone during the demo

Suggested demo script (3 to 4 minutes)

1. A farmer (a teammate in the role) speaks to the app; no typing at any point.
2. Show the pre-walked field polygon with two crop patches and their areas.
3. Upload a diseased leaf photo; the app speaks the diagnosis and pins it on the map.
4. Ask "should I water today?"; the answer cites rain forecast and soil values.
5. Ask "what will I earn from wheat?"; show the profit range from live prices.
6. End with Fasal Score and one line on impact.

Prepare in advance: a pre-recorded GPS walk for a backup demo, 5 to 10 real disease photos, cached weather and price data in case the network fails, and one native-language speaker to test the voice.

Risks, limitations and future scope

The biggest risks are wrong advice reaching a farmer and voice failing in a noisy field; both have simple mitigations.

Risk	Impact	Mitigation
Misdiagnosis of disease	Wrong or wasted treatment, crop loss	Show confidence, ask for better photos, escalate low-confidence or severe cases to a KVK or agri officer
Unsafe chemical advice	Health and environmental harm	Prefer organic and IPM options first; chemical advice only from verified label guidance
Speech errors from noise or dialect	Wrong action taken	Read back and confirm every save; allow "no, repeat"
GPS inaccuracy (phone GPS is often several metres off)	Wrong area, wrong profit	Drop poor fixes, smooth the path, let the farmer adjust on the map
Missing or stale soil data	Poor fertiliser advice	Soil Health Card OCR, regional defaults labelled "estimated", prompt for fresh test yearly
Price source changes or fails	Profit and score wrong	Official feed first, cached last value with date shown
Poor connectivity	App unusable	Offline queue, cached advice, voice-call fallback
Privacy of boundaries and photos	Loss of trust	Voice consent, per-farmer storage, delete on request
API cost at scale (vision and speech)	Budget overrun	Compress images, cache repeated answers, run a cheap on-device filter before the vision call

Future scope

- Satellite crop-health maps (NDVI) to spot stress before it is visible
- Tie Fasal Score to crop insurance and credit eligibility with farmer consent
- Market linkage: find the best nearby mandi or buyer and suggest when to sell

- Pest and disease outbreak alerts shared across neighbouring farms
- Government scheme finder with voice-guided application help
- Farmer groups and FPO dashboards for bulk advice and selling